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Association between poverty status and self care practices among diabetic patients in Morocco

Aicha El Hanafi^{1*} , Safiya Mahlaq² and Mohamed Lmejjati¹

*Correspondence:

Aicha El Hanafi

aicha13.hanafi@gmail.com

¹REGNE Research Laboratory,
Faculty of Medicine and Pharmacy
of Agadir, Ibn Zohr University,
Agadir, Morocco

²Laboratory of Biostatistics, Clinical
Research and Epidemiology
(LBRCE), Faculty of Medicine and
Pharmacy of Rabat, Mohammed V
University, Rabat, Morocco

Abstract

Background Diabetes is a chronic and progressive disease that affects individuals' daily lives by causing both acute and chronic complications. Its control largely depends on self-management by the patient, but precarious socioeconomic conditions limit its effectiveness.

Objectives This study aims to determine the impact of living in poverty on self-care practices among diabetic patients and identify the practices that are impaired by poverty.

Methods A cross-sectional analytical study was conducted among diabetic patients in the province of Agadir Ida Outane, Morocco. Self-care practices were measured via the Moroccan Arabic version of the SDSCA questionnaire. Logistic regression was employed to predict the relationship between poverty status and self-care practices after adjusting for potential confounding variables. The standard significance threshold was set at 0.05. Data processing and analysis were carried out via Jamovi statistical analysis software, version (2.3.28).

Results A total of 377 confirmed diabetic patients who were receiving treatment were included in the study. The results of multiple logistic regressions show that poverty is associated with poor diabetes self-care practices (ORa = 1.72; 95% CI [1.01–2.94]; $p = 0.047$). The self-care dimensions significantly linked to poverty status are physical exercise ($p = 0.005$), blood glucose monitoring ($p < 0.001$), and foot care ($p < 0.001$).

Conclusion Implementing appropriate measures to strengthen diabetes self-management among patients in precarious socioeconomic conditions is essential.

Keywords Poverty, Socioeconomic level, Self-care, Self-management, Diabetes, Morocco

1 Introduction

In developing countries, diabetes is a chronic disease that generates considerable medical costs because of its high prevalence, prolonged course, and associated micro- and macrovascular complications. The United Nations recognized it as a global health threat



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comparable in severity to infectious epidemics, making it the first noncommunicable disease [1].

In Morocco, the epidemiological situation of diabetes remains alarming. More than 2 million adults aged 18 and over are affected, and half of them are unaware of their condition [2]. Diabetes is a leading cause of mortality in the country, responsible for over 12,000 deaths each year. In addition, complications associated with this pathology contribute to an additional 32,000 deaths [3].

Studies on diabetes management have long highlighted the importance of patient involvement. Active participation helps individuals reach their glycemic and HbA1c targets [4]. This role involves a series of self-care practices, including regular blood glucose monitoring, adjusting insulin doses on the basis of food intake and physical activity, and managing or treating hypoglycemia. Maintaining a strict therapeutic routine is often challenging, especially for financially vulnerable patients with limited access to care and essential resources [5]. Several studies have highlighted that socioeconomic disparities strongly influence diabetes management. These disparities affect many aspects, including patients' individual capabilities [6]. In low-income populations, self-care engagement is often limited due to financial constraints and misconceptions about the disease and its treatment requirements [7].

In short, self-care practices are a central element in diabetes management. Optimizing them is essential to enable patients to better manage their disease [8]. Socioeconomic inequalities in diabetes self-management are well documented. However, studies on secular trends in these disparities remain limited. This underscores the need for a deeper understanding of how poverty affects self-management among diabetic patients. The present study aimed, first, to describe self-care practices among diabetic patients and, second, to assess the impact of an unfavorable socioeconomic situation, specifically poverty, on these practices.

2 Methods

2.1 Research design

The research methodology employed in this study is observational, cross-sectional, and analytical. The study was conducted in 2023 and focused on diabetic patients living in the province of Agadir Ida Outanane, located in the Souss Massa region of southern Morocco. Its main objective is to examine the effect of poverty on self-care practices among diabetic patients.

This study followed the (Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) study design recommendations.

2.2 Participants

Participants were selected through accidental sampling from ambulatory diabetic patients attending urban and rural primary care facilities. Inclusion criteria included adults aged 18 and over, diagnosed with type 1 or type 2 diabetes, and under treatment for at least 1 year. Exclusion criteria were: women with gestational diabetes, patients not

attending primary care facilities, those with mental illness affecting intellectual capacity, and patients of foreign nationality.

The website CheckMarket [9], a sample size calculator, was used to obtain a sample of 377 people based on a total diabetic population of 18,149, followed up at the primary care facilities in the province of Agadir Ida Outanane, with a margin of error of 5% and a confidence interval of 95%. Data were collected from participants during their follow-up consultations or during treatment dispensation in specialized chronic disease units. Selection was made from six health centers that were statistically representative of diabetic patients under their care until the required minimum sample size was reached.

2.3 Data collection and measurement instruments

Data were collected on sociodemographic, clinical, therapeutic, and psychological characteristics commonly referenced in the literature on disease self-management [10]. These variables were gathered using a structured questionnaire developed by the researchers:

Sociodemographics: age (<55 years and $\geq$ 55 years); sex (male, female); place of residence (urban, rural); marital status (married, single, divorced, widowed); educational level (illiterate, primary, secondary, university); employment status (active, inactive, retired); poverty status (yes, no); health insurance (without, obligatory health insurance (AMO) solidarity, AMO).

Clinical: duration of disease ($\leq$ 10 years and $>$ 10 years); most recent HbA1c level according to three modalities (<7% 'optimal', between 7 and 9% 'suboptimal', and $>$ 9% 'high-risk'), comorbidity (yes, no), chronic complications (yes, no), obesity (yes, no), and smoking status (yes, no).

The following therapeutics were used: type of treatment (insulin therapy, oral treatment, or oral treatment + insulin therapy); treatment compliance (low, medium, or high); frequency of medical consultations per year (less than 3 months, every 3 months, or more than 3 months); use of traditional treatment (yes or no); and having received information about diabetes from healthcare professionals (yes or no).

The perception of the disease, whether low or high, reflects its psychological dimension. According to the literature, a positive perception can motivate patients to adopt appropriate behaviors and plays a key role in treatment success [11]. Thus, among the contextual factors that influence the self-care integration process are patients' lived experiences with the disease [12].

The fundamental variable of this study is the poverty situation, measured by the criterion of relative poverty in Morocco established by Morocco's High Commission for Planning (HCP) (a person living in poverty with an income threshold equivalent to 50% of the median per capita income) [13]. This approach, which is based on household income, is the most widely used approach in most national and international surveys.

Treatment adherence is measured using the Arabic version of the Morisky, Green, and Levine (MGL) adherence scale, adapted and validated in 2022 by Awwad et al. [14]. The adapted version demonstrated strong reliability and validity, with psychometric properties comparable to the original English version. The instrument includes four questions

and was designed to detect medication nonadherence in patients with chronic diseases, particularly diabetes [14]. It is based on a simplified scoring system, assigning a value of 1 for a “Yes” response and 0 for a “No” response to each item. A low score indicates better compliance. Patients' total scores can be classified into three categories: high adherence (response “yes” to 0 items), moderate adherence (response “yes” to 1 or 2 items), and low adherence (response “yes” to 3 or 4 items) [15].

In Morocco, health insurance is regulated by Law 65–00, which defines two primary schemes. The first is AMO, a compulsory plan that covers employees in both public and private sectors, as well as retirees and their dependents. The second is (Ramed), a medical assistance program designed for people living in poverty and vulnerable areas [16]. In 2022, a social protection reform introduced AMO Solidarity to replace Ramed. The goal was to offer more effective coverage for households with limited financial resources. During 2023, a transition period allowed some beneficiaries to retain temporary access to Ramed while AMO Solidarity was gradually rolled out [17]. Since this study took place during the transitional year of 2023, the available medical coverage options at that time included the following: (no coverage, RAMED/AMO solidarity, and AMO).

Clinical data on treatment and the latest HbA1c level are extracted from patients' follow-up records. Obesity status is determined using the body mass index (BMI), calculated with Quetelet's formula: (weight (kg)/height²). According to World Health Organization (WHO) standards, obesity is defined as a BMI ≥ 30 kg/m² [18].

Chronic complications included conditions such as eye conditions, heart conditions, kidney conditions, and foot conditions (neuropathy and diabetic foot). Comorbidities are defined as other associated chronic diseases. They can be either declared directly by the patient or recorded in the patient's diary when the information is available.

Disease self-care is measured using the Moroccan Arabic version of the SDSCA, “Summary of Diabetes Self-Care Activities”, adapted and validated by Adarmouch et al. [19]. It was adapted from the revised version by Toobert et al. in [20] and includes 11 items assessing various aspects: the antidiabetic regimen (general diet, specific dietary recommendations), physical exercise, blood glucose testing, foot care, and smoking [20]. The Moroccan Arabic version of the SDSCA includes 8 items, as items 3 and 4 posed problems during adaptation and showed weak psychometric properties during validation. This 8-item version was used to assess four dimensions of diabetes self-care: diet, exercise, glycemic control, and foot care [19]. The items measure the frequency of performance of self-management activities for each domain over the last 7 days. Responses range from 0 to 7 (number of days). Item 2 differs from the other items in that it asks about average weekly frequency over the past month. We calculate the score for each domain by averaging the values of the two items that constitute it [19].

Good self-care practices were defined using a threshold of 5 days per week for each dimension, in line with recommendations from the literature [21, 22]. As in previous studies [23, 24], the overall score was calculated by summing the mean scores of each

measured dimension. The final score ranges from 0 (lowest) to 28 (highest). The participants were categorized as having “high” or “low” self-care practices, using the median of the final score. The validity and reliability of the Moroccan Arabic version of the SDSCA were proven.

Regarding illness perception, it was assessed using the Arabic version of the Brief Illness Perception Questionnaire (B-IPQ), translated and validated by Saarti et al. (2016), which showed satisfactory psychometric properties [11]. Originally developed in English-speaking countries, it is a widely adopted nine-item scale [25].

The scale measures patients' cognitive and emotional representations of their illness. Five of the items assess cognitive representations of the disease: consequences (item 1), chronology (item 2), personal control (item 3), treatment control (item 4), and identity (item 5). Two of the items assess emotional representations: worry (item 6) and emotions (item 8). One item evaluated the comprehensibility of the disease (item 7). Causal representation is assessed through open-response item 9, adapted from the IPQ-R. It invites patients to list the three main causes they associate with their disease [25]. Responses are scored on a scale from 0 to 10. For items 1, 2, 5, 6, and 8, a score of 0 indicates a good perception of the disease, and a score of 10 indicates a bad perception of it. For items 3, 4, and 7, a score of 0 indicates a poor perception of the disease, and a score of 10 indicates a good perception of it [11, 25].

The scores of the 8 items were summed after reversing the scores for items 3, 4, and 7 to obtain a total score ranging from 0 to 80. The Brief IPQ tool primarily measures negative illness perceptions, as reflected in the questions. However, items 3, 4, and 7 assess positive perceptions of the illness, which is why their scores were reversed before calculating the overall score. A high total score indicates that the participant feels threatened by their diabetes [24].

The assessment of perceived causes is coded separately and is not discussed in this article. Although this item does not contribute to the overall adherence score. Disease level perceptions was categorized into two groups, with “high” and “low” negative disease perceptions falling above and below the median, respectively. The Cronbach's alpha for this scale in the present study is 0.7, which indicates acceptable reliability.

Data collection from diabetic individuals was carried out via the aforementioned instruments through face-to-face interviews conducted in a neutral and nonsuggestive manner due to the high rates of low educational attainment. The average time required for each patient was approximately 15–20 min.

2.4 Statistical analysis

Descriptive data are presented as means and standard deviations (SD) for normally distributed continuous variables, or as medians and interquartile ranges (IQR) for those not following a normal distribution.

Table 1 Sociodemographic characteristics of the participants (n = 377)

Sociodemographic characteristics	Participant n = 377 (n%)
Gender	
Male	147 (39)
Female	230 (61)
Age (years) (median & IQR)	55 [45–64]
< 55 ans	180 (47.7)
> = 55 ans	197 (52.2)
Living area	
Rural	98 (26)
Urban	279 (74)
Marital status	
Single	41 (10.9)
Married	257 (68.2)
Divorced	25 (6.6)
Widowed	54 (14.3)
Occupational status	
Active	125 (33.2)
Inactive	227 (60.2)
Retired	25 (6.6)
Medical coverage	
None	75 (19.9)
Ramed/AMO 'solidarity'	144 (38.2)
AMO	158 (41.9)
Poverty status	
Yes	150 (39.8)
No	227 (62.2)
Educational level	
Unschooling	174 (46.2)
Primary school	74 (19.6)
Secondary school	93 (24.7)
University	36 (9.5)

IQR: interquartile range, Ramed: medical assistance plan, AMO: obligatory health insurance; HbA1c: glycated hemoglobin, N: number, and %: percentage

A univariate logistic regression analysis was then carried out:

Initially, the focus was on analyzing the dimensions of self-care practices in connection with poverty status, aiming to delve deeper into the nature of this relationship. Second, we examined the relationships between the overall self-care score and all the other sociodemographic, clinical, and therapeutic variables in this study.

Finally, a multivariate logistic regression model was constructed to analyze the overall self-care score. It included key sociodemographic, clinical, and therapeutic factors identified in the univariate analyses. The objective was to examine the relationship between poverty status and self-care practices, while adjusting for other covariates.

All the statistical analyses were performed via jamovi software version 2.3.28, with a significance level of 0.05 for all the statistical analyses.

Table 2 Clinical, therapeutic and psychological characteristics of the participants (n = 377)

Clinical, therapeutic and psychological characteristics	Participant n = 377 (n%)
Duration of diabetes	
< = 10	196 (52)
> 10	181 (48)
HBAC1 level (%)	
< 7	56 (14.9)
7–9	265 (70.3)
> 9	56 (14.9)
Chronic complications	
Yes	187 (49.6)
No	190 (50.4)
Comorbidity	
Yes	152 (40.3)
No	225 (59.7)
Obesity status	
Yes	27 (7.2)
No	350 (92.8)
Smoking status	
Yes	35 (9.3)
No	342 (90.7)
Adherence to treatment	
Low	25 (6.6)
Moderate	117 (31)
High	235 (62.3)
Frequency of visits	
Less than 3 months	6 (1.6)
Every 3 months	175 (46.4)
More than 3 months	196 (52)
Treatment type	
Insulin alone	164 (43.5)
Oral antidiabetics	151 (40.1)
Insulin + Oral antidiabetics	62 (16.4)
Traditional treatment	
Yes	80 (21.2)
No	297 (78.8)
having received information about diabetes from healthcare professionals	
Yes	342 (90.7)
No	35 (9.3)
Negative illness perception	
Final score (median and IQR)	47 [39–52]
Low	187 (49.6)
High	190 (50.4)

Table 3 Self-care activities among participants (n = 377)

Dimensions	Items	All participants median [IQR]	All participants N(%)		Poverty status median [IQR]		P value	Cronbach's α
			Low	High	Yes	No		
Diet	02	5 [3.5–7]	152 (40.3)	225 (59.7)	5 [3.5–6.5]	6 [3–7]	0.162	0.91
Physical activity	02	3.5 [2–6]	247 (65.5)	130 (34.5)	3.5 [2–5]	4 [1.5–6]	0.005	0.72
Glycemic control	02	1 [0–2]	327 (86.7)	50 (13.3)	0 [0–1.38]	2 [0–3.25]	<0.001	0.97
Foot care	02	2 [0–5]	282 (74.5)	96 (25.5)	1 [0–3]	3 [0–5.50]	<0.001	0.86
Overall self-care		12 [9–16]	179 (47.5)	198 (52.5)	10.5 [8–14]	13.5 [9.25–18.3]	<0.001	–

IQR: interquartile range, N: number, and %: percentage

3 Results

Among the 377 participants in this study, 61% were women. The age distribution revealed that 52.2% of the participants were ≥ 55 years old. In terms of educational level, 46.2% had no formal education, whereas only 9.5% had a university level of education. In terms of provenance, 74% were from urban areas. The study also revealed that 68.2% of the participants were married and that 60.2% had no professional activity. A total of 41.9% had AMO medical coverage. Finally, 39.8% of the participants were living in poverty. (Table 1).

Forty-eight percent of the participants had been diagnosed for more than 10 years, 70.3% had an HbA1c value between 7 and 9%, and 62.3% had a high level of treatment compliance. A total of 49.6% of the patients had chronic complications, and 40.3% had a comorbidity. In addition, 43.5% of the patients were treated with insulin, whereas 40.1% were treated orally. A total of 7.2% and 9.3% of the large minority adolescents were obese and smokers, respectively (Table 2).

With respect to access to healthcare, only 46.4% of the participants regularly consulted their doctor every 3 months, whereas 90.7% had received information on diabetes from healthcare professionals. Finally, a significant proportion (21.2%) used conventional treatment, and half (50.4%) had a high negative perception of diabetes. (Table 2).

According to the percentages of self-care practices, more than half of the participants (52.5%) had high overall self-care scores. The majority showed high levels in the diet domain (59.7%). In contrast, a greater proportion of participants reported low levels of blood glucose (86.7%), foot care (74.5%) and exercise (65.5%). Cronbach's alpha for the diet, physical activity, blood glucose testing, and foot care scores was 0.91, 0.72, 0.97, and 0.86, respectively, demonstrating the excellent reliability of the Moroccan Arabic version of the SDSCA used in our study (Table 3).

Additionally, we used univariate logistic regression to identify the associations between various self-care practices and living in poverty. In fact, a statistically significant difference was observed between groups living in poverty and those not living in poverty, particularly in terms of glycemic control (OR = 9.28; 95% CI [3.26–26.37]; $p < 0.001$), foot care (OR = 3.84; 95% CI [2.19–6.74]; $p < 0.001$), physical activity (OR = 1.90; 95% CI [1.21–2.99]; $p = 0.005$) and overall self-care (OR = 2.33; 95% CI [1.526–3.545]; $p < 0.001$). (Table 4).

To examine the association between poverty and low overall self-care, it was essential to adjust for potential confounding variables. Therefore, a univariate analysis was conducted on the overall self-care score and the relevant socioeconomic, clinical, and therapeutic variables included in the study. The results identified 14 significant variables:

Table 4 Univariate logistic regression for different aspects of self-care practices in relation to poverty status

Practice	Diet			Physical activity			Glycemic control			Foot care			Overall self-care		
	OR	IC 95%	P	OR	IC 95%	P	OR	IC 95%	P	OR	IC 95%	P	OR	IC 95%	P
Poverty Yes-No	1.35	0.89–2.05	0.162	1.90	1.21–2.99	0.005	9.28	3.26–26.37	<0.001	3.84	2.19–6.74	<0.001	2.33	1.53–3.54	<0.001

The estimate represents the log odds of low vs. high practices

Table 5 Univariate logistic regression for the different predictors of overall self-care

	Univariate logistic regression		
	Self care		
	OR	IC	P
Poverty status:			
Yes–No	2.33	1.53–3.54	< 0.001
Gender:			
Female–Male	1.55	1.02–2.36	0.039
Age (years)	1.03	1.01–1.04	0.002
Living area:			
Rural—Urban	1.21	0.76–1.92	0.415
Marital status:			
Single	Ref	Ref	Ref
Married	1.72	0.86–3.42	0.125
Divorced	1.51	0.55–4.20	0.424
Widowed	3.03	1.30–7.06	0.010
Occupational status:			
Active	Ref	Ref	Ref
inactive	2.38	1.51–3.74	< 0.001
retired	1.27	0.53–3.07	0.593
Medical coverage:			
None	1.82	1.04–3.17	0.035
Ramed/AMO 'solidarity'	1.64	1.04–2.59	0.034
AMO	Ref	Ref	Ref
Educational level:			
Unschooling	Ref	Ref	Ref
Primary school	1.59	0.90–2.80	0.108
Secondary school	0.41	0.24–0.69	< 0.001
University	0.07	0.02–0.25	< 0.001
Duration of diabetes:			
< = 10–> 10	0.84	0.56–1.26	0.40
HBAC1 level (%)			
< 7	Ref	Ref	Ref
7–9	1.38	0.76–2.49	0.288
> 9	3.52	1.61–7.67	0.002
Chronic complications:			
Yes–No	2.01	1.33–3.03	< 0.001
Comorbidity:			
Yes–No	2.12	1.39–3.22	< 0.001
Obesity status:			
Yes–No	1.42	0.64–3.12	0.38
Smoking status:			
Yes–No	0.71	0.35–1.45	0.35
Adherence to treatment:			
Low	2.63	1.09–6.34	0.031
Moderate	1.18	0.75–1.83	0.474
High	Ref	Ref	Ref
Frequency of visits:			
Less than 3 months	0.29	0.03–2.50	0.258
Every 3 months	Ref	Ref	Ref
More than 3 months	1.68	1.12–2.54	0.013
Treatment type:			
Insulin alone	0.70	0.45–1.09	0.11
Oral antidiabetics	Ref	Ref	Ref
Insulin + Oral antidiabetics	1.28	0.71–2.32	0.42

Table 5 (continued)

	Univariate logistic regression		
	Self care		
	OR	IC	P
Traditional treatment:			
Yes–No	2.03	1.23–3.36	0.006
Negative illness perception:			
High–Low	2.46	1.62–3.72	< 0.001

Table 6 Multivariate logistic regression for different predictors of overall self-care

Variable	Multivariate logistic regression		
	Overall self-care		
	ORa	IC 95%	P
Educational level			
Unschooling	Ref	Ref	Ref
Primary school	2.19	1.13–4.24	0.020
Secondary school	0.77	0.37–1.59	0.476
University	0.13	0.03–0.54	0.005
Traditional treatment (Yes)	1.88	1.03–3.41	0.039
Negative illness perception (High)	1.84	1.10–3.08	0.020
Poverty status (Yes)	1.72	1.01–2.94	0.047

ORa: Adjusted odds ratio; IC: Confidence interval

p-value in bold: significant value with a standard significance level set at 0.05

sex, age, professional activity, marital status, education level, medical coverage, chronic complications, comorbidities, HbA1c level, treatment compliance, frequency of medical visits, use of traditional treatments, negative illness perceptions, and poverty status (Table 5).

In a multivariate analysis, four variables were found to be associated with low overall self-care among participants: level of education, use of traditional treatment, high negative perception of disease, and poverty status (ORa = 1.72; 95% CI [1.01–2.94]; $p = 0.047$) (Table 6).

4 Discussion

Self-care practices are essential for diabetic patients to manage their condition and prevent complications. Effective diabetes control is challenging without a clear understanding of patients' current self-care behaviors [26]. We examined the strength of the association between poverty and self-care practices among diabetic patients, and found significant correlations indicating that individuals living in poverty were more likely to engage in poor self-care behaviors.

To the best of our knowledge, this study is the first quantitative research conducted in Morocco to explicitly examine the relationship between poverty and self-care practices among patients with diabetes. Unlike many studies that approach diabetes from a biomedical or behavioral standpoint, this research emphasizes the direct influence of socio-economic conditions on patients' daily self-management behaviors. It offers a contextualized and integrated perspective on diabetes, taking into account the material and social realities that shape patient experiences. This framework moves beyond strictly medical explanations, proposing systemic and community-based solutions that are better aligned with the needs of vulnerable populations.

To further quantify this relationship, multivariate logistic regression analysis revealed a significant association between poverty and low diabetes self-care (ORa = 1.72; 95% CI [1.01–2.94]; $p = 0.047$). This relationship was demonstrated after adjusting for potential confounding covariates involved in the study. This result is supported by other research, notably that of Nelson et al. (2016), which highlighted lower adherence to self-care among individuals with low income. This trend is attributed mainly to financial constraints and misconceptions about diabetes and its management [7]. Reshma et al. (2021), also reported a direct link between diabetes self-management and financial factors such as family income. Thus, diabetes is more influential in disadvantaged social environments [27]. These findings are supported by other studies, including those by Vaezi et al. (2018) [28] and Davari et al. (2014) [29], which confirmed a significant association ($p < 0.05$) between self-care scores and family income among individuals with diabetes. In summary, diabetes management is inadequate because of economic insecurity, which has a negative effect on the quality of life of the individuals concerned [30].

Across the different dimensions analyzed, low self-care practices were significantly associated with poverty status, independent of other covariates. For physical exercise, the odds ratio was (OR = 1.90; 95% CI [1.21–2.99]; $p = 0.005$), with mean scores differing between poor and non-poor groups (3.5 [2–5] vs. 4 [1.5–6]). Glycemic control showed a stronger association, with (OR = 9.28; 95% CI [3.26–26.37]; $p = < 0.001$) and median scores of (0 [0–1.38] vs. 2 [0–3.25]). Foot care was also significantly linked to poverty, with (OR = 3.84; 95% CI [2.19–6.74]; $p = < 0.001$) and scores of (1 [0–3] vs. 3 [0–5.50]) for the poor group compared to the non-poor group.

Another study revealed important differences in physical exercise ($F = 8.22$), blood glucose testing ($F = 3.08$), and foot care ($F = 22.38$) between upper- and upper-middle-class patients and those in lower social class groups [31].

Previous research incorporating socioeconomic status into the analysis of factors influencing self-care practices in diabetic patients has reached similar conclusions. In fact, adherence to physical exercise was more common in patients with higher socioeconomic status [32]. However, low-income patients are less likely to exercise [33]. Patients from lower socioeconomic backgrounds tend to be more physically inactive, mainly because they cannot afford to take part in club sports or similar activities [34].

In relation to glycemic control, unfavorable socioeconomic status, represented by poverty, remains a significant factor in this practice. Studies in Morocco have confirmed these results [21, 35, 36]. Patients with higher incomes tend to perform blood glucose tests more frequently than those with lower incomes, a disparity largely driven by the cost of care. Fewer than half of diabetic patients (46.8%) conducted blood glucose tests at home, and only 18% did so daily. Financial barriers to accessing home monitoring devices, such as glucometers and blood glucose strips, significantly hinder effective diabetes self-management [37, 38].

With respect to the foot care dimension, an integrative literature review revealed that low socioeconomic status had an impact on foot care behaviors among diabetic patients. An improvement in foot care practices was observed among patients with higher incomes [39]. Furthermore, in the study by Rondhianto et al. (2023), income level was the only sociodemographic factor that significantly affected foot care habits ($p = 0.019$). Low-income respondents reported poor foot care behavior [40]. Similarly,

low socioeconomic status has been identified as a risk factor for inadequate foot care practices, which in turn increases the likelihood of diabetic foot complications [41].

This relationship between poverty status and the adoption of self-care practices suggests that, beyond medical treatment, addressing the social determinants of health, including poverty status, is essential to effective diabetes management.

4.1 Policy recommendations

The results of this study highlight a pressing need to reshape diabetes care strategies with a stronger emphasis on social equity and inclusive access. Poverty emerges not merely as a lack of material resources, but as a powerful social determinant that shapes patients' perceptions, behaviors, and capacity to engage in effective self-care. These insights carry significant implications for both practical interventions and public health policy:

- Tailored health education and support: Public health programs must develop educational methods that are culturally and economically appropriate, taking into account literacy levels and socioeconomic status. Community health workers and peer educators can play a vital role in bridging the gap between medical advice and lived realities.
- Incorporating of a social approach in diabetes management: Engaging social workers and civil society organizations is essential to mitigate barriers faced by vulnerable populations in accessing essential healthcare resources. Targeted interventions such as the provision of glucometers to facilitate routine glycemic monitoring can enhance self-management capacities and promote equitable health outcomes.
- Advancing health equity through structural socioeconomic policy interventions: This includes reinforcing universal health coverage to ensure comprehensive access to care, instituting minimum income guarantees to reduce financial vulnerability, and mitigating geographic disparities in healthcare accessibility. Such multisectoral strategies are essential to promote social justice and improve population health across diverse socioeconomic contexts.

4.2 Study strengths and limitations

The main strengths of our study include the use of a validated Moroccan version of the SDSCA, which was designed to increase the accuracy of self-care practice assessments, as well as the inclusion of diabetes patients from both urban and rural areas in primary care settings across Morocco. However, certain limitations must be considered. First, conducting the study within a single province using accidental sampling and a relatively small sample size may limit the generalizability of the findings to the broader population of Moroccan patients with diabetes. Second, the assessment of relative poverty is based solely on household income, which may not capture all aspects of socioeconomic status. Third, the reliance on self-reported data for assessing self-care practices introduces the possibility of recall bias and social desirability bias, which may have influenced participants' responses and affected the accuracy of the findings. Finally, the study did not consider potential confounding factors, such as disease knowledge and social support.

5 Conclusion

Our results suggest that efforts to improve self-care should focus on tackling socioeconomic inequalities. It is essential to develop strategies that help people from disadvantaged socioeconomic groups adopt healthier diabetes self-management practices and to raise awareness among diabetic patients about the importance of self-management in reducing the occurrence of complications and improving the quality of life of people with diabetes.

Abbreviations

Ramed	The medical assistance plan
AMO	Obligatory health insurance
HbA1c	Glycated hemoglobin
B-IPQ	Brief illness perception questionnaire
SDSCA	Summary of diabetes self-care activities

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Author contributions

E.A. was responsible for the study design, data collection, analysis, interpretation and manuscript writing. M.S. participated in the data analysis and interpretation. L.M. participated in the study design, data analysis and interpretation, edited the manuscript and presented the final version. All the authors have read and approved the final version and approved the submission of the manuscript.

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Data availability

The data utilized and/or examined in this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with ethical standards and with full respect for the dignity and rights of the participants. Free and informed consent was obtained from all participants after the interviewer explained the objectives of the study and its voluntary nature. Participants were assured that their personal information would remain confidential and anonymous. The study complied with the principles outlined in the Declaration of Helsinki regarding research involving human subjects. Ethical approval for this study was obtained from the Ethics Committee for Biomedical Research of the Faculty of Medicine and Pharmacy of Rabat Mohammed V University (File number 63/23).

Consent for publication

Not applicable. This study does not contain any person's data in any form (including individual details, images, or videos). All data were analyzed in an aggregated and anonymized format.

Informed consent

Informed consent was obtained from all individual participants included in the original survey.

Conflict of interest

The authors declare no competing interests.

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